Multiple Anchors and RIS-Aided Localization Method in Complex NLOS Environments

Zhenqian Wu, Youming Li, Xiaolong Zhang, Xiangpei Meng, Xinrong Lv, and Yonghong Wu

***Abstract*—This article addresses the localization problem of wireless sensor networks (WSNs) in complex non line-of-sight (NLOS) environments where multiple regions are separated by the blockages. Reconfigurable intelligent surface (RIS) and ultrawideband (UWB) technology are exploited to form a novel multiple anchors and RIS-aided localization (MARL) method, in which anchors and RISs are deployed in pairs in each separated region. The method consists of two stages: 1) a coarse stage and 2) a fine stage. In the coarse stage, each RIS is dynamically adjusted in accordance with predesigned conjugate orthogonal configuration sequence to extract the direct or reflected path components effectively from all anchors during each UWB pulse period. A coarse position of the target is then determined based on the time-delay estimation of the extracted path components, while NLOS bias is assumed to exist in each path. In this stage, the region where the target exists is also identified. In the fine stage, a reflection response vector estimation process is initiated by adjusting the RIS located in the identified region, which is used to formulate a minimum mean square error (MMSE) problem. The solution of the MMSE problem is calibrated with the path coefficient generated from the previously acquired coarse estimate. Then, a reflection vector that contains the true position of the target is obtained. Finally, an accurate estimate is achieved by using gradient descent search algorithm. Simulation results show that the proposed MARL method is a cost-effective scheme for target localization in NLOS environments of WSNs.**

***Index Terms*—Non line-of-sight (NLOS), reconfigurable intelligent surface (RIS), ultrawideband (UWB), wireless sensor networks (WSNs).**

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I. Introduction

WIRELESS localization plays an essential role in location-based services of wireless sensor networks (WSNs) [1], including military applications [2], environment surveillance [3], cyber–physical systems [4], early warning and rescue operation [5], etc. Traditionally, WSNs localization is achieved by transmitting wireless signals, such as Wi-Fi, Bluetooth, ultrawideband (UWB), or other signals, between sensors and anchors when GPS systems are not accessible in indoor or dense urban areas [6]. More recently, efforts have been made to enhance localization accuracy by utilizing millimeter-wave (mmWave) or terahertz signals in WSNs [7]. Nevertheless, the efficacy and dependability of localization are persistently impeded by the non line-of-sight (NLOS) channel conditions, wherein signal blockages by obstructions pose a formidable challenge [8]. Therefore, the pivotal inquiry remains on actualizing a dependable localization method amidst unfavorable propagation environments for WSNs.

Multianchor localization is a traditional method that makes full use of link information from multiple predeployed anchors in NLOS environments. A common localization approach typically begins by identifying NLOS and line-of-sight (LOS) links. Subsequently, various parameters like time-delay, angle of arrival (AOA), angle of departure (AOD), or received signal strength (RSS) of the identified LOS links are utilized to perform localization [9]. Although this kind of approach can yield good results, NLOS link identification is a difficult and time-consuming problem [10]. Another approach involves explicitly expressing NLOS bias in each link and optimizing localization based on prior knowledge of NLOS bias, such as its distribution [11] or upper bound [12], which are scenario-dependent and difficult to obtain. It is worth noting that both approaches have shown notable benefits in UWB-based localization due to the high-multipath resolution of UWB signals [13]. However, their performance heavily depends on the number and proportion of LOS links [14]. Therefore, a large number of anchors are necessary to ensure localization accuracy in NLOS environments, resulting in increased cost, deployment difficulty, and algorithm complexity.

Recently a promising technology for addressing the localization challenge under NLOS channel conditions is the use of reconfigurable intelligent surface (RIS), which is composed of massive passive programmable reflecting elements. RIS can efficiently control the electromagnetic wave, including phase and amplitude, to enable intelligent reconfiguration of the radio environment [15]. The reflected paths via RIS provide

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additional and essential channel information, such as time-delay, AOA, AOD, and RSS. Currently, research on RIS-aided NLOS localization mainly focuses on partially obstructed NLOS scenes where the direct path is blocked and the reflected path via RIS is unobstructed, or small open areas that can be fully covered by a single base station [16], [17], [18], [19]. Elzanaty et al. [16] and Zhang et al. [17] investigated a localization scenario where a single base station and a single RIS are deployed. The localization was performed in the near-field condition by using a curved wavefront model to express the reflected path in a mmWave multiple-input multiple-output (MIMO) orthogonal frequency division multiplexing (OFDM) setup. Han et al. [18] addressed a joint localization and near-field channel reconstruction problem for large-scale RIS-assisted massive MIMO systems operating in time division duplex (TDD) transmission mode. The channel between the unique base station and the RIS is reconstructed after determining the positions of multiple sensors. In [19], the time-delay of each reflected path via each RIS's element was estimated by sending a sequence of OFDM pilot symbols. This method can achieve centimeter-level localization accuracy, even in scenarios where parts of the reflected paths were interfered with or blocked. However, an exceptionally large scale RIS must be deployed around the entire perimeter of the localization field to ensure coverage and reliability.

The wireless localization in NLOS environments is a challenging research topic. In recent years, the RIS-aided communication technology has provided new solutions for addressing the wireless localization problem. However, in expansive scenes encompassing multiple meeting rooms or large gymnasiums consisting of multiple venues, obstructions between rooms occur frequently, giving rise to complex NLOS localization environments. Currently, there is a lack of in-depth research on localization in this scenario.

In this work, we develop a novel multiple anchors and RIS-aided localization (MARL) method for localization in complex NLOS scene by fully utilizing RIS's capability of adjusting the reflected path. In a complex NLOS scene containing several separate regions, we deploy one anchor and one RIS pair in each region. Intraregion path is guaranteed to be LOS, while inter-region path is likely to be NLOS. Hence, we perform coarse localization based on the inter-region paths and fine localization based on the intraregion paths. Due to the reflection of RISs, each anchor receives a bunch of reflected path signals, leading to a significant enhancement of time-delay-based localization compared to scenarios without RISs. Therefore, we first obtain a coarse position in a time difference of arrival (TDOA) problem after isolating all direct and reflected path components, which is achieved by reconfiguring the RIS in each UWB pulse period. Besides, the presence region of the target is identified by searching for the signal with the highest strength. In the fine stage, we employ a reflection response vector estimation process in LOS condition by reconfiguring the RIS within the identified region. An accurate position estimate is finally achieved through minimum mean square error (MMSE) optimization and gradient descent search. The versatility of our MARL method allows it to adapt to various complex NLOS scenes. The main contributions of this work are summarized below.

1) In order to address the challenges posed by complex NLOS environments, we incorporate multiple anchors and multiple RISs, constructing a comprehensive channel model that encompasses direct channels as well as reflection channels from RISs. We formulate the localization problem in complex NLOS environments and propose a MARL-based localization method.
2) In the coarse stage, we propose a signal correlation operation aiming at extracting the direct and reflected path components that are mixed in the received signal of the anchors. Additionally, we provide design strategy on the configuration sequences of RISs to facilitate the extraction of these path components.
3) We perform a fine localization by determining the reflection response vector of the RIS. This reflection vector is obtained based on the estimation of the LOS reflected path's gain with the coarse position estimate in the previous stage.

The remainder of this article is organized as follows. Section II introduces the system model. Section III proposes the coarse localization step in which a time difference of arrival (TDOA) localization is performed to generate a rough estimate. Section IV provides the fine-grained localization step. Simulation results are depicted in Section V. Finally, conclusions are drawn in Section VI.

Notations: We use $\bar{A}$, A^{T}, and A^{H} to denote the conjugate, the transpose, and the conjugate transpose of vector A, respectively. $\otimes$ is the Kronecker product operation. $[\cdot]_{n,m}$ denotes parameters or signals in the reflected path via the mth RIS to the nth Anchor. $[\cdot]_{m>n}$ denotes parameters of the second hop from the mth RIS to the nth Anchor in the reflection path. $[\cdot]^t$ denotes the parameter or signal in the tth UWB pulse period. $\{\cdot\}$ denotes a time sequence of variables. We use $\mathbf{1}_K$ to denote the 1-by-K vector of ones and $\mathbf{I}_K$ to denote the K-dimensional identity matrix.

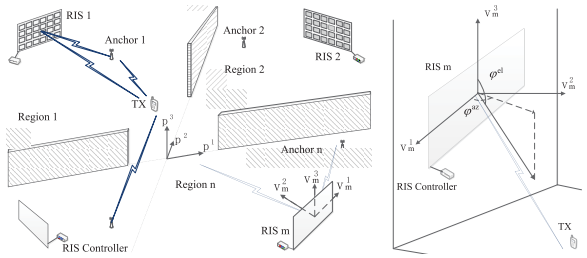
II. System Model

A. Localization System

Consider a sizeable wireless localization working field (denoted by $\mathbb{B}$) divided into M separate regions by walls, building partitions, or other obstructions, which is indicated by $b(b = 1, 2, \ldots, M)$. As shown in Fig. 1(a), we deploy M single antenna anchors and M RISs, and arrange these anchors and RISs in pairs in each separate region, denoted by $A_n(n = 1, 2, \ldots, M)$ and $R_m(m = 1, 2, \ldots, M)$, respectively. All the RISs are centrally controlled. All the anchors function as receivers (RX) for training signals. After deploying an anchor-RIS pair in a specific region, it is reasonable to assume that communication within that region occurs under LOS conditions. However, communication between different regions is possible in NLOS conditions.

The target acts as the transmitter (TX) with an unknown position that need to be estimated. The TX sends single-tone UWB signals $s^t(\eta) = p(\eta)e^{j2\pi f_c \eta}$, where t indicates a

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(a) (b)

Fig. 1. (a) Wireless localization system in the large working field. (b) Elevation and azimuth angles of a vector.

sending fragment in a pulse-sending sequence, f_c is the carrier frequency in mmWave band and $p(\eta)$ is the baseband pulse.

We consider a 3-D scenario, and establish a general coordinate system $\mathcal{G}$ with the center of the working field $\mathbb{B}$ as the origin. All the RIS planes are arranged vertically w.r.t. the ground, and their azimuth is fixed. For further convenience, RIS's coordinate system is created with the center of RIS as the origin, denoted as follows:

$$\mathcal{R}_m = \left[v_m^1, v_m^2, v_m^3\right]^{\mathrm{T}} \qquad (m = 1, \ldots, M). \tag{1}$$

Fig. 1(b) shows the definition of the azimuth and elevation of the reflected signal in each RIS's coordinate system. We assume the rotation matrix from the general coordinate system to the RIS R_m's coordinate system is known, which is denoted as $\mathbf{R}_{\mathcal{G}\to\mathcal{R}_m}$.

All the RISs are identical in size and structure, each consisting of an M_r by M_c uniform planar array with an interelement distance equal to half the wavelength of the carrier frequency, denoted by d. We suppose the configuration coefficient of the element in each RIS can be independently controlled to change the phase of the impinging electromagnetic wave in $0 \sim 2\pi$ phase shifts.

B. Channel Model

As one anchor is concerned, the propagation channel is composed of the direct channel from the TX and multiple reflected channels from M RISs. The direct channel coefficient of anchor A_n from the TX is expressed as

$$\mathbf{H}_n^{(d)} = \varepsilon_n^{\mathrm{LOS}} \alpha_n^1 e^{j2\pi f_c(\tau_n^1 + t_0)} + \sum_{l=2}^{L_n} \alpha_n^l e^{j2\pi f_c(\tau_n^l + t_0)} \tag{2}$$

where $\varepsilon_n^{\mathrm{LOS}} \in \{0, 1\}$ is the indicator of blockage of the LOS direct path, and α_n^l and τ_n^l are channel gain and time-delay of the lth path, respectively. The parameters (α_n^1, τ_n^1) characterize the LOS direct path. According to [19], $\alpha_n^1 = (\sqrt{G_{\mathrm{T}} G_R P_t}\lambda/4\pi\tau_n^1 c)$, where G_{T} and G_R are the antenna gain of the transmitter and receiver, respectively, λ is the wavelength of the carrier wave, P_t is the transmitted power, c is the electromagnetic wave velocity.

In this work, we assume that all the anchors are time-synchronized, but the TX and the anchors are asynchronous. Hence, t_0 in (2) denotes the clock bias between the TX and the anchors and needs to be estimated.

The reflected channel coefficient of anchor A_n via RIS R_m is expressed as

$$\mathbf{H}_{n,m}^{(r)} = \mathbf{G}_m^{\mathrm{T}} \mathrm{diag}(\mathbf{\Omega}_m) \mathbf{G}_{m>n} \tag{3}$$

where $\mathbf{G}_m$ is the first hop channel coefficient from the transmitter TX to RIS R_m, $\mathbf{G}_{m>n}$ is the second hop channel coefficient from RIS R_m to anchor A_n, and $\mathbf{\Omega}_m = [\Omega_{m,1}, \ldots, \Omega_{m,K}]^{\mathrm{T}}$ is the configuration vector referring to the set of phase shifts applied to RIS R_m, where $K = M_r M_c$ is the number of elements in each RIS.

We rewrite (3) in the following form:

$$\mathbf{H}_{n,m}^{(r)} = \mathbf{G}_m^{\mathrm{T}} \mathrm{diag}(\mathbf{G}_{m>n}) \mathbf{\Omega}_m \tag{4}$$

where $\mathbf{G}_m^{\mathrm{T}} \mathrm{diag}(\mathbf{G}_{m>n})$ is called the cascade channel that represents the reflected channel characteristics.

We consider far-field communication when the communication range is beyond the Fraunhofer distance. The plane wavefront assumption is used for the reflected channel model of RIS. Hence, the reflected channel coefficient is described as follows:

$$\mathbf{H}_{n,m}^{(r)} = \mathbf{D}_{n,m}^1 \mathbf{\Omega}_m + \sum_{l=2}^{L_{n,m}} \mathbf{D}_{n,m}^l \mathbf{\Omega}_m \tag{5}$$

where

$$\mathbf{D}_{n,m}^1 = \varepsilon_{n,m}^{\mathrm{LOS}} \delta_{n,m}^1 \mathbf{a}^{\mathrm{T}}(\theta_m) \mathrm{diag}(\mathbf{a}(\phi_{m>n})) \tag{6}$$

$$\mathbf{D}_{n,m}^l = \delta_{n,m}^l \mathbf{a}^{\mathrm{T}}\left(\theta_m^l\right) \mathrm{diag}\left(\mathbf{a}(\phi_{m>n}^l)\right) \tag{7}$$

where $\varepsilon_{n,m}^{\mathrm{LOS}} \in \{0, 1\}$ denotes the blockage indicator of the reflected path from the TX to anchor A_n via RIS R_m and $\delta_{n,m}^1$ is the complex gain of the LOS reflected path, with the definition $\delta_{n,m}^1 = (\sqrt{G_{\mathrm{T}} P_t}\lambda/4\pi\tau_m^1 c)[\sqrt{G_R}\lambda/4\pi\tau_{m>n}^1 c] e^{j2\pi f_c(\tau_m^1 + \tau_{m>n}^1 + t_0)}$, where τ_m^1, $\tau_{m>n}^1$ is the time-delay of the first and second hop in the LOS reflected path, respectively. $\theta_m = [\theta_m^{az}, \theta_m^{el}]^{\mathrm{T}}$ in (6) denotes the AOA in azimuth and elevation from the TX to RIS R_m, and $\phi_{m>n} = [\phi_{m>n}^{az}, \phi_{m>n}^{el}]^{\mathrm{T}}$ is the AOD from RIS R_m to anchor A_n.

The item $\mathbf{a}^{\mathrm{T}}(\theta_m)\mathrm{diag}(\mathbf{a}(\phi_{m>n}))$ represents the reflection response vector from TX to anchor A_n via RIS R_m. The definition of the function $\mathbf{a}(\psi)$ is

$$\mathbf{a}(\psi) = \mathbf{a}_r(\kappa_1) \otimes \mathbf{a}_c(\kappa_3) \tag{8}$$

and $\mathbf{a}_r(\kappa_1)$, $\mathbf{a}_c(\kappa_3)$ is defined, respectively, as follows:

$$\mathbf{a}_r(\kappa_1) = e^{j\beta_r}\left[1, e^{j\frac{2\pi d}{\lambda}\kappa_1}, \ldots, e^{j\frac{2\pi d}{\lambda}(M_r-1)\kappa_1}\right]^{\mathrm{T}} \tag{9}$$

$$\mathbf{a}_c(\kappa_3) = e^{j\beta_c}\left[1, e^{j\frac{2\pi d}{\lambda}\kappa_3}, \ldots, e^{j\frac{2\pi d}{\lambda}(M_c-1)\kappa_3}\right]^{\mathrm{T}} \tag{10}$$

where $\beta_r = -(\pi d/\lambda)(M_r - 1)\kappa_1$ and $\beta_c = -(\pi d/\lambda)(M_c - 1)\kappa_3$. Here

$$\begin{aligned}\kappa &= [\kappa_1, \kappa_2, \kappa_3] \\ &= \left[\sin\left(\psi^{el}\right)\cos\left(\psi^{az}\right), \sin\left(\psi^{el}\right)\sin\left(\psi^{az}\right), \cos\left(\psi^{el}\right)\right]\end{aligned} \tag{11}$$

is the wavenumber vector.

The parameters $(\delta_{n,m}^l, \theta_m^l, \phi_{m>n}^l)$ in (7), and $\mathbf{D}_{n,m}^l$ in (5) characterize the multipath part that is incident to or reflected from RIS R_m, respectively.

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In this article, we assume that the antenna gain G_{T} and G_R are both unknown. Hence, the definition of the gain coefficient $\delta_{n,m}^1$ in (5) cannot be exploited directly for signal processing. However, if the configuration vector $\mathbf{\Omega}_m$ is adjusted to satisfy $\mathbf{a}^{\mathrm{T}}(\theta_m)\mathrm{diag}(\mathbf{a}(\phi_{m>n}))\mathbf{\Omega}_m = K$, the LOS gain coefficient $\delta_{n,m}^1$ can be estimated directly with the reflected component received by anchor A_n, which is narrated in Section IV. So we call the following configuration vector as the gain optimal beam $\mathbf{\Omega}_m^\star$:

$$\mathbf{\Omega}_m^\star = \left\{\mathbf{a}^{\mathrm{T}}(\theta_m)\mathrm{diag}(\mathbf{a}(\phi_{m>n}))\right\}^{\mathrm{H}} \tag{12}$$

which can be calculated with the true position of the TX.

III. Coarse Localization Stage

In NLOS scenarios with multiple anchors and RISs deployed, the problem of localization is very challenging, as the received information is a mixture of direct and reflected wave components. In this section, we discuss how to identify the direct and reflected path components from the received information by adjusting the configuration vector of RISs in signal pulse period. Based on the extracted component of each path, the TDOA localization is performed to acquire the coarse estimate.

A. Extraction of Each Component

For a time-delay-based localization system with M anchors without the assistance of RIS, we can only obtain M time-delay parameters of direct paths. However, in the proposed MARL system, M^2 reflected paths are created as a result of the reflection of each RIS to each anchor. If each signal component of the direct path and the reflected path is separated, a total of $M + M^2$ time-delay parameters will be obtained, significantly enhancing the accuracy of localization. Therefore, in this section, we extract the signals by utilizing the phase control capability of RIS. Within each pulse period, the configuration vector of each RIS is adjusted based on a predesigned sequence.

The signal y_n^t captured at anchor A_n during the tth pulse period is represented as follows:

$$y_n^t = y_n^{d,t} + \sum_{m=1}^{M} y_{n,m}^{r,t} + w_n^t, \quad t = 1, 2, \ldots, \mathrm{T} \tag{13}$$

where $y_n^{d,t}$ represents the direct component transmitted from the TX to anchor A_n, while $y_{n,m}^{r,t}$ denotes the reflected component that bounces off RIS R_m and reaches anchor A_n, and $w_n^t \sim \mathcal{CN}(0, \sigma^2)$ denotes the additive white Gaussian noise (AWGN) with power σ^2.

Based on the channel coefficient in (2) and (5), the direct and reflected component can be, respectively, represented as follows:

$$y_n^{d,t} = \mathbf{H}_n^{(d)} s^t(\eta) \tag{14}$$

$$y_{n,m}^{r,t} = \mathbf{D}_{n,m}^1 \mathbf{\Omega}_m^t s^t(\eta) + \sum_{l=2}^{L_{n,m}} \mathbf{D}_{n,m}^l \mathbf{\Omega}_m^t s^t(\eta) \tag{15}$$

where $\mathbf{\Omega}_m^t \in \mathbb{C}^{K\times 1}$ denotes the configuration vector of RIS R_m in the period of the tth pulse, for $t = 1, 2, \ldots, \mathrm{T}$.

Since the transmitted UWB signal $s^t(\eta)$ is known at the receiver, without loss of generality, in the remainder of this article, we will set $s^t(\eta) = 1 \ \ \forall t$.

The direct component in (14) remains constant because the direct channel is assumed to be time-invariant channel during the short period of the T-length UWB pulse. On the other hand, the reflection components in (15) rely on the RIS configuration vector. By combining all received signals y_n^t, it can isolate the direct component by removing the reflection component. We have the following combination of all $y_n^t (t = 1, \ldots, \mathrm{T})$ which are obtained by substituting (14) and (15) into (13)

$$\sum_{t=1}^{\mathrm{T}} y_n^t = \sum_{t=1}^{\mathrm{T}} y_n^{d,t} + \sum_{t=1}^{\mathrm{T}} w_n^t + \sum_{t=1}^{\mathrm{T}} \sum_{m=1}^{M} \left(\mathbf{D}_{n,m}^1 \mathbf{\Omega}_m^t + \sum_{l=2}^{L_{n,m}} \mathbf{D}_{n,m}^l \mathbf{\Omega}_m^t \right). \tag{16}$$

According to (16), the reflected component can be removed by using a well-designed RIS configuration vector sequence. The M configuration vector sequences $\{\mathbf{\Omega}_m^t\}$ $(m = 1, 2, \ldots, M)$ are generalized in the following equation:

$$\mathbf{\Omega}_m^t = \omega_m^t \mathbf{\Omega}_m^B, (m = 1, 2, \ldots, M), (t = 1, 2, \ldots, \mathrm{T}) \tag{17}$$

where $\mathbf{\Omega}_m^B \in \mathbb{C}^{K\times 1}$ is an initial configuration vector and $\omega_m^t \in \mathbb{C}^{1\times 1}$ is a phase shift factor at the tth pulse period.

In (17), all the elements of the RIS undergo the same phase shift from the previous pulse period to the next.

By substituting $\mathbf{\Omega}_m^t$ into (16), we have

$$\sum_{t=1}^{\mathrm{T}} y_n^t = \sum_{t=1}^{\mathrm{T}} y_n^{d,t} + \sum_{t=1}^{\mathrm{T}} w_n^t + \sum_{m=1}^{M} \left(\mathbf{D}_{n,m}^1 + \sum_{l=2}^{L_{n,m}} \mathbf{D}_{n,m}^l \right) \mathbf{\Omega}_m^B \sum_{t=1}^{\mathrm{T}} \omega_m^t. \tag{18}$$

We observe that if the summation of the phase shift factor sequence is assumed to be zero, i.e.,

$$\sum_{t=1}^{\mathrm{T}} \omega_m^t = 0, (m = 1, 2, \ldots, M) \tag{19}$$

then we have the estimated direct component $\hat{y}_n^d$ from the TX to anchor A_n in the following equation after neglecting the noise item:

$$\hat{y}_n^d = \frac{1}{\mathrm{T}} \sum_{t=1}^{\mathrm{T}} y_n^t, (n = 1, 2, \ldots, N). \tag{20}$$

To extract the reflected component, we multiply the received signal y_n^t by the conjugate of each RIS configuration vector $\{\mathbf{\Omega}_{m'}^t\}, m' = 1, 2, \ldots, M$, sum them up for all pulse periods of the T-length sequence, and then substitute the RIS configuration vector expression (17) into $\{\mathbf{\Omega}_{m'}^t\}$, we can obtain

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$$u_{n,m,m'} = \sum_{t=1}^{\mathrm{T}} y_n^t \bar{\mathbf{\Omega}}_{m'}^t = \mathbf{H}_n^{(d)} \bar{\mathbf{\Omega}}_{m'}^B \sum_{t=1}^{\mathrm{T}} \omega_{m'}^t + \sum_{m=1}^{M} \left(\mathbf{D}_{n,m}^1 \mathbf{\Omega}_m^B \bar{\mathbf{\Omega}}_{m'}^B \sum_{t=1}^{\mathrm{T}} \omega_m^t \bar{\omega}_{m'}^t + \sum_{l=2}^{L_{n,m}} \mathbf{D}_{n,m}^l \mathbf{\Omega}_m^B \bar{\mathbf{\Omega}}_{m'}^B \sum_{t=1}^{\mathrm{T}} \omega_m^t \bar{\omega}_{m'}^t \right) + \bar{\mathbf{\Omega}}_{m'}^B \sum_{t=1}^{\mathrm{T}} w_n^t \bar{\omega}_{m'}^t, \quad m' = 1, 2, \ldots, M. \tag{21}$$

Based on the condition that all the phase shift factor sequences are conjugate orthogonal, that is

$$\sum_{t=1}^{\mathrm{T}} \omega_m^t \bar{\omega}_{m'}^t = 0 \quad \forall m \neq m', \left(m, m' = 1, 2, \ldots, M\right) \tag{22}$$

then we have

$$u_{n,m,m'} = \begin{cases} \bar{\mathbf{\Omega}}_{m'}^B \sum_{t=1}^{\mathrm{T}} w_n^t \bar{\omega}_{m'}^t, & \text{if } m \neq m' \\ T y_{n,m}^r \bar{\mathbf{\Omega}}_{m'}^B + \bar{\mathbf{\Omega}}_{m'}^B \sum_{t=1}^{\mathrm{T}} w_n^t \bar{\omega}_{m'}^t, & \text{if } m = m'. \end{cases}$$

Hence, we have the following expression for the estimated reflected component which bounces off RIS R_m and arrives at anchor A_n:

$$\hat{y}_{n,m}^r = \frac{1}{\mathrm{T}} \frac{\mathbf{1}_K \left(\sum_{t=1}^{\mathrm{T}} y_n^t \bar{\mathbf{\Omega}}_m^t \right)}{\mathbf{1}_K \bar{\mathbf{\Omega}}_m^B}, (n, m = 1, 2, \ldots, M). \tag{23}$$

The phase shift factor sequences $\{\omega_m^t\}$ satisfying (19) and (22) can be generated from Hadamard matrix [15]. When $\mathrm{T} \geq M + 1$, the orthogonality property of these configuration vector sequences $\{\mathbf{\Omega}_m^t\}(m = 1, \ldots, M)$ can be met.

B. TDOA Localization After Time-Delay Estimation

Now we can estimate the localization based on time-delay information by using the direct path component in (20) and reflected path component in (23). However, the localization problem can not be solved directly from solving the maximum likelihood estimator of the unknown time-delay parameters $[\tau_n^1, \tau_m^1, \tau_{m>n}^1, t_0]$ in (2) and (5) as the LOS indicators $\varepsilon_n^{\mathrm{LOS}}$ and $\varepsilon_{n,m}^{\mathrm{LOS}}$ cannot be identified. To address this challenge, we first estimate the time-delay and then formulate a TDOA optimization problem.

Time-delay estimation can be accomplished by using techniques, such as cross-correlation function, threshold value, or peak detection, as proposed in [21]. Let $\hat{\tau}_n^d$ denote the time-delay estimation of the direct component $\hat{y}_n^d$. According to the channel coefficient in (2), the time-delay estimation $\hat{\tau}_n^d$ is expressed as

$$\hat{\tau}_n^d = \tau_n^1 + t_0 + \gamma_n + n_n \tag{24}$$

where τ_n^1 is the time-delay of the LOS direct path, γ_n represents the NLOS bias that occurs in NLOS direct path, and n_n is zero mean Gaussian variable.

Let $\hat{\tau}_{n,m}^r$ denote the time-delay estimation of the reflected component $\hat{y}_{n,m}^r$. According to the channel coefficient in (5), the time-delay estimation $\hat{\tau}_{n,m}^r$ is expressed as

$$\hat{\tau}_{n,m}^r = \tau_m^1 + \tau_{m>n}^1 + t_0 + \gamma_{n,m} + n_{n,m} \tag{25}$$

where τ_m^1 and $\tau_{m>n}^1$ is the time-delay of the first and second hop in the LOS reflected path, respectively, $\gamma_{n,m}$ represents the NLOS bias that occurs in NLOS reflected path, and $n_{n,m}$ is zero mean Gaussian variable.

Multiplying both sides of (24) and (25) by the electromagnetic wave velocity and expressing the LOS time-delay in terms of the position of TX, anchors and RISs, we obtain the following geometric equalities:

$$c\hat{\tau}_n^d = \|p_S - p_{A_n}\| + ct_0 + c\gamma_n + cn_n \tag{26}$$

$$c\hat{\tau}_{n,m}^r = \|p_S - p_{R_m}\| + \|p_{R_m} - p_{A_n}\| + ct_0 + c\gamma_{n,m} + cn_{n,m} \tag{27}$$

where p_S, p_{A_n}, and p_{R_m} denotes the position of the TX, anchor A_n, and RIS R_m, respectively.

We notice that the term $\|p_{R_m} - p_{A_n}\|$ in (27) is known because the anchors and RISs are predeployed. Hence, (27) can be rewritten as

$$c\hat{\tau}_{n,m}^r - \|p_{R_m} - p_{A_n}\| = \|p_S - p_{R_m}\| + ct_0 + c\gamma_{n,m} + cn_{n,m}. \tag{28}$$

We formulate the following TDOA optimization problem according to the MMSE principle:

$$\begin{aligned} \min_{p_S, r_{n1}, r_{(n,m)1}} \quad & (f_1 + f_2) \\ \text{s.t.} \quad f_1 &= \sum_{n=2}^{M} \left(\rho_{n1} + \|p_S - p_{A_n}\| - \|p_S - p_{A_1}\| + r_{n1} \right)^2 \\ f_2 &= \sum_{n=2}^{M} \sum_{m=1}^{M} (\rho_{(n,m)1} + \|p_S - p_{R_m}\| - \|p_S - p_{A_1}\| + r_{(n,m)1})^2 \end{aligned} \tag{29}$$

where $\rho_{n1} = c\hat{\tau}_n^d - c\hat{\tau}_1^d$, $\rho_{(n,m)1} = c\hat{\tau}_{n,m}^r - c\hat{\tau}_{1,m}^r - \|p_{R_m} - p_{A_n}\| + \|p_{R_m} - p_{A_1}\|$, $r_{n1} = \gamma_n - \gamma_1$, $r_{(n,m)1} = \gamma_{n,m} - \gamma_{1,m}$ and p_{A_1} is the position of the anchor with the minimum one of all time-delay estimation of direct components.

Problem (29) refers to a nonconvex TDOA localization problem that involves mitigating NLOS bias. This problem can be addressed by employing the method proposed in [22], which transforms the original problem into a semi-definite programming problem. The position of the TX $\hat{p}_S$ is then estimated, and a suboptimal beam $\hat{\mathbf{\Omega}}_m^\star$ is generated from (12), which can be used to estimate the LOS reflected path coefficient in Section IV.

However, we notice that problem (29) is ill-conditioned, and the accuracy of the solution $\hat{p}_S$ is affected by the NLOS biases r_{n1} and $r_{(n,m)1}$, which is unknown nuisance parameters for problem (29). Hence, the solution $\hat{p}_S$ can only be used as a coarse position estimate of the target.

Notice that the communication in a separate region is in LOS condition according to the system deployment in Section II, we can improve the localization accuracy based on the LOS components reflected by the RIS that is located in the same region with the target. Therefore, we perform the fine localization stage in Section IV. But now we need to identify the region where the target is located for further operation. Since the anchor in the same region with the target

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has the greatest direct wave component, we determine the region indictor by solving the following maximum searching problem:

$$\hat{n} = \arg\max_{n} |\hat{y}_n^d| \tag{30}$$

where $\hat{y}_n^d$ is the direct component received by anchor A_n, calculated in (20).

IV. Fine Localization Stage

In this section, we try to obtain an accurate estimate of the TX's position based on the LOS reflected channel provided by RIS. We define $\mathbf{v}_S = \mathbf{a}^{\mathrm{T}}(\theta_{\hat{n}})\mathrm{diag}(\mathbf{a}(\phi_{\hat{n}>\hat{n}}))$ as the reflection response vector of RIS $R_{\hat{n}}$, where $\hat{n}$ is the region number identified in search problem (30) based on the extracted direct component in the coarse stage. As the definition shows, the vector $\mathbf{v}_S$ is characterized by the position of TX. It can obtain a precise estimate of the TX's position if $\mathbf{v}_S$ is accurately determined. Hence, we initiate the following reflection response vector estimation process to obtain $\mathbf{v}_S$. The reflected channel's complex gain is approximated by employing the gain optimal beam $\hat{\mathbf{\Omega}}_{\hat{n}}^{\star}$ derived from (12) along with the coarse position $\hat{p}_S$, which is then used to determine $\mathbf{v}_S$. Finally, a gradient descent algorithm is employed to search for the fine position.

A. Reflection Response Vector Estimation

We merely activate RIS $R_{\hat{n}}$ within Region $\hat{n}$, deactivating all the other RISs outside of this region. We collect the wireless measurements at $A_{\hat{n}}$ during a $(K+1)$-length UWB pulse sequence, while RIS $R_{\hat{n}}$ is reconfigured by using randomly generated configuration vectors in the first K pulse periods, and is reconfigured by using $\hat{\mathbf{\Omega}}_{\hat{n}}^{\star}$ in the last pulse period.

Neglecting the multipath part in (5) for simplicity, we have the reflected component model as follows:

$$y_{\hat{n},\hat{n}}^{r,t} = \delta_{\hat{n},\hat{n}}^1 \mathbf{v}_S \mathbf{\Omega}_{\hat{n}}^t + w_{\hat{n},\hat{n}}^{r,t} \tag{31}$$

where $\mathbf{v}_S = \mathbf{a}^{\mathrm{T}}(\theta_{\hat{n}})\mathrm{diag}(\mathbf{a}(\phi_{\hat{n}>\hat{n}})) \in \mathbb{C}^{1\times K}$ denotes the reflection response vector of RIS $R_{\hat{n}}$, $\delta_{\hat{n},\hat{n}}^1$ is the complex gain of the LOS reflected path from TX to anchor $A_{\hat{n}}$ via RIS $R_{\hat{n}}$, and $w_{\hat{n},\hat{n}}^{r,t}$ is a zero mean Gaussian noise following the distribution $\mathcal{N}(0, \sigma_{\hat{n}}^2)$.

The reflected components of $K+1$ UWB pulse periods are estimated as $\hat{y}_{\hat{n},\hat{n}}^{r,t} = y_{\hat{n}}^t - \hat{y}_{\hat{n}}^d (t = 1, \ldots, K+1)$, where $y_{\hat{n}}^t$ is the signal received at anchor $A_{\hat{n}}$, $\hat{y}_{\hat{n}}^d$ is the direct component obtained in the coarse localization stage in (20).

To obtain the reflection response vector $\mathbf{v}_S$, a new MMSE problem is formulated by using the first K reflected components as follows:

$$\begin{aligned} \min_{\mathbf{v}_S} \quad & \|\hat{\mathbf{Y}} - \delta_{\hat{n},\hat{n}}^1 \mathbf{\Gamma}_{\hat{n}} \mathbf{v}_S^{\mathrm{T}}\|^2 \\ \text{s.t.} \quad & |v_{Sk}| = 1 \quad (k = 1, 2, \ldots, K) \end{aligned} \tag{32}$$

where $\hat{\mathbf{Y}} = [\hat{y}_{\hat{n},\hat{n}}^{r,1}, \ldots, \hat{y}_{\hat{n},\hat{n}}^{r,K}]^{\mathrm{T}}$ is the observed vector of reflected components and $\mathbf{\Gamma}_{\hat{n}} = [\mathbf{\Omega}_{\hat{n}}^1, \ldots, \mathbf{\Omega}_{\hat{n}}^K]^{\mathrm{T}}$ is the configuration matrix consisting of K configuration vectors over K pulse, v_{Sk} is the kth entry in $\mathbf{v}_S$ and the constraint $|v_{Sk}| = 1 (k = 1, 2, \ldots, K)$ is satisfied according to the definition of $\mathbf{v}_S$.

We express the globally optimal solution of (32) as follows:

$$\hat{\mathbf{v}}_S = \frac{1}{\delta_{\hat{n},\hat{n}}^1} \hat{\mathbf{Y}}^{\mathrm{T}} \mathbf{\Gamma}_{\hat{n}} \left(\mathbf{\Gamma}_{\hat{n}}^{\mathrm{T}} \mathbf{\Gamma}_{\hat{n}}\right)^{-1} \tag{33}$$

where $\mathrm{rank}(\mathbf{\Gamma}_{\hat{n}}) = K$ is normally satisfied because the configuration vector of RIS $R_{\hat{n}}$ is randomly generated.

However, the complex gain of the reflected path $\delta_{n,\hat{n}}^1$ in (33) is still unknown. An approximate value of $\delta_{n,\hat{n}}^1$ is estimated by using the reflected component of the last training period when RIS $R_{\hat{n}}$ is configured with the gain optimal beam $\hat{\mathbf{\Omega}}_{\hat{n}}^{\star}$, which is obtained from (12) with the estimate $\hat{p}_S$ in the coarse localization stage. Neglecting the noise item in (31), we have

$$\delta_{\hat{n},\hat{n}}^1 = \frac{1}{\mathbf{v}_S \hat{\mathbf{\Omega}}_{\hat{n}}^{\star}} \hat{y}_{\hat{n},\hat{n}}^{r,t^{\star}}, \quad \left(\mathbf{\Omega}_{\hat{n}}^{t^{\star}} = \hat{\mathbf{\Omega}}_{\hat{n}}^{\star}\right) \tag{34}$$

where $\hat{y}_{\hat{n},\hat{n}}^{r,t^{\star}}$ is the reflected component in the last pulse period when RIS $R_{\hat{n}}$ is configured as $\hat{\mathbf{\Omega}}_{\hat{n}}^{\star}$.

Referring to the proof in Appendix A, $\mathbf{v}_S \hat{\mathbf{\Omega}}_{\hat{n}}^{\star}$ is a real number for any given $\hat{\mathbf{\Omega}}_{\hat{n}}^{\star}$ satisfying the inequality $\mathbf{v}_S \hat{\mathbf{\Omega}}_{\hat{n}}^{\star} \le K$. We approximately estimate $\delta_{\hat{n},\hat{n}}^1$ by substituting $\mathbf{v}_S \hat{\mathbf{\Omega}}_{\hat{n}}^{\star}$ with K

$$\hat{\delta} = \frac{1}{K} \hat{y}_{\hat{n},\hat{n}}^{r,t^{\star}}, \quad \left(\mathbf{\Omega}_{\hat{n}}^{t^{\star}} = \hat{\mathbf{\Omega}}_{\hat{n}}^{\star}\right) \tag{35}$$

which satisfies the following relation:

$$\delta_{\hat{n},\hat{n}}^1 = \frac{1}{k} \hat{\delta} \tag{36}$$

where $k = ([\mathbf{v}_S \hat{\mathbf{\Omega}}_{\hat{n}}^{\star}]/K)$ is a constant, denoting the estimated proportional deviation of $\delta_{\hat{n},\hat{n}}^1$. Thus, we rewrite the estimate of $\mathbf{v}_S$ as follows:

$$\hat{\mathbf{v}}_S = \frac{1}{\hat{\delta}} \hat{\mathbf{Y}}^{\mathrm{T}} \mathbf{\Gamma}_{\hat{n}} \left(\mathbf{\Gamma}_{\hat{n}}^{\mathrm{T}} \mathbf{\Gamma}_{\hat{n}}\right)^{-1}. \tag{37}$$

According to the constraint $|v_{Sk}| = 1 (k = 1, 2, \ldots, K)$ in problem (32), a unitization step is required to eliminate the estimated proportional deviation k. Hence, a final estimation of the reflection response vector $\mathbf{v}_S^*$ is obtained by unitizing each entry in $\hat{\mathbf{v}}_S$ as follows:

$$v_{Sk}^* = \frac{\hat{v}_{Sk}}{|\hat{v}_{Sk}|}, \quad (k = 1, 2, \ldots, K). \tag{38}$$

B. Fine Position Search

According to the definition in (8), the reflection response vector $\mathbf{v}_S$ is a function of the TX's true position. We formulate the following problem to solve the TX's position by approaching $\mathbf{v}_S^*$:

$$\begin{aligned} \min_{p_S} \quad & \|\mathbf{a}^{\mathrm{T}}(\theta_{\hat{n}})\mathrm{diag}(\mathbf{a}(\phi_{\hat{n}>\hat{n}})) - \mathbf{v}_S^*\|^2 \\ \text{s.t.} \quad & \theta_{\hat{n}} = \left[\theta_{\hat{n}}^{az}, \theta_{\hat{n}}^{el}\right]^{\mathrm{T}} \\ & \sin\left(\theta_{\hat{n}}^{el}\right) = b_s^{el}(p_S) \\ & \cos\left(\theta_{\hat{n}}^{az}\right) = b_c^{az}(p_S) \\ & \cos\left(\theta_{\hat{n}}^{el}\right) = b_c^{el}(p_S) \end{aligned} \tag{39}$$

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Algorithm 1 Gradient Descent Localization Search

Input: the coarse estimate $\hat{p}_S$ in the coarse localization stage, the reflected vector $\mathbf{v}_S^*$, the preset parameter ϵ, the preset step size η

Output: p_S^*

1: Generate N_{p_S} initial position p_x randomly around the coarse estimate $\hat{p}_S$
2: **for all** $x \in (1, 2, \ldots, N_{p_S})$ **do**
3: **repeat**
4: $p_x^{i+1} = p_x^i - \eta\left(\frac{\partial L(\kappa_1,\kappa_3)}{\partial p_x}\big|_{p_x=p_x^i}\right)$
5: **until** $(p_x^{i+1} - p_x^i)^2 \le \epsilon$
6: **end for**
7: $\hat{x} = \arg\min_x \left(L(\kappa_1, \kappa_3)\big|_{p_x}\right), x \in (1, 2, \ldots, N_{p_S})$
8: **return** $p_S^* = p_{\hat{x}}$

where $b_c^{el}(p_S)$, $b_s^{az}(p_S)$, and $b_c^{el}(p_S)$ are geometric expressions in terms of p_S in RIS $R_{\hat{n}}$'s coordinate, and $\mathbf{a}(\phi_{\hat{n}>\hat{n}})$ is known vector calculated in (8)–(11) with $p_{R_{\hat{n}}}$ and $p_{A_{\hat{n}}}$.

It is worth noting that the positions of Anchor $A_{\hat{n}}$, RIS $R_{\hat{n}}$, and the TX in (39) are expressed in the RIS $R_{\hat{n}}$'s coordinate system, which are converted from the global coordinate system by the known rotation matrix $\mathbf{R}_{\mathcal{G}\to\mathcal{R}_m}$. For simplicity, we omit the coordinate transformation and do not change the notation of positions in this section.

Problem (39) is nonconvex and very difficult to deal with. We apply gradient descent algorithm to search the optimal solution from multiple randomly generated initial position. First, we rewrite problem (39) as follows:

$$\begin{aligned} \min_{p_S} \quad & L(\kappa_1, \kappa_3) \\ \text{s.t.} \quad & L(\kappa_1, \kappa_3) = \left(\mathbf{v}(\kappa_1, \kappa_3) - \mathbf{v}_S^*\right)\left(\mathbf{v}(\kappa_1, \kappa_3) - \mathbf{v}_S^*\right)^{\mathrm{H}} \\ & \mathbf{v}(\kappa_1, \kappa_3) = (\mathbf{a}_r(\kappa_1) \otimes \mathbf{a}_c(\kappa_3))^{\mathrm{T}} \mathrm{diag}(\mathbf{a}(\phi_{\hat{n}>\hat{n}})) \\ & \kappa_1 = b_s^{el}(p_S) b_c^{az}(p_S) \\ & \kappa_3 = b_c^{el}(p_S) \end{aligned} \tag{40}$$

where $\mathbf{a}_r(\kappa_1)$ and $\mathbf{a}_c(\kappa_3)$ are defined in (9) and (10), respectively.

Then, we apply the search Algorithm 1 to get the final solution p_S^*.

The gradient expression of $[\partial L(\kappa_1, \kappa_3)/\partial p_x]$ in Algorithm 1 is given in Appendix B. This gradient descent localization search is guaranteed to find the optimal solution of problem (40) by the effectiveness analysis in Appendix C.

V. Numerical Results

In this section, we numerically study the performance of the proposed MARL scheme in the complex NLOS environments.

A. Settings of Numerical Experiment

We assume that the MARL system is deployed in indoor scenarios. The complex NLOS environment is a square area, where four separate regions exist due to the partition of walls or barriers. Each region in this area is deployed with one anchor in the center and one RIS near the anchor. The TX is randomly located in this area with unknown position $p_S = [x, y, 0]$. The rest system parameters are listed in Table I, where the rotation matrix $\mathbf{R}_{\mathcal{G}\to\mathcal{R}_m}$ is omitted because it can be expressed in the azimuth of RISs' plane in the general coordination.

TABLE I
Simulation Parameters

Parameter	Symbol	Value
Carrier frequency	f_c	28GHz
TX antenna gain	G_T	6 dB
RX antenna gain	G_R	2 dB
TX-RX clock uncertainty	T_a	$50 \sim 100$ ns
Transmission power	P_t	-5 dBm
Noise power	σ^2	$-55 \sim -115$ dBm
Number of paths	L	10
Path arrival rate	T_τ	8 ns
Ratio of LOS paths	ρ_L	$25\% \sim 95\%$
Width of a single UWB pulse	τ_s	2 ns
Length of UWB pulses	T	$8 \sim 128$
Number of RIS elements	K	$M_c \times M_r$ $M_c, M_r = 4 \sim 12$
RIS element's spacing	d	$\lambda/2$
Position of Anchors	p_{A_n}	$(5,5,5), (-5,5,5)$ $(5,-5,5), (-5,-5,5)$
Position of RISs	p_{R_m}	$(8,6,5), (-8,6,5)$ $(8,-6,5), (-8,-6,5)$
Azimuth of RISs' plane	φ	$\frac{5\pi}{4}, \frac{7\pi}{4}, \frac{3\pi}{4}, \frac{\pi}{4}$

An asynchronous network is simulated with synchronized anchors, but a clock offset t_0 exists at the TX. In addition, t_0 is modeled according to the clock uncertainty T_a in Table I. ρ_L represents the proportion of LOS channel, including direct paths from anchors and reflected paths from RISs. The LOS indicators $\varepsilon_n^{\mathrm{LOS}}$ and $\varepsilon_{n,m}^{\mathrm{LOS}}$ in (2) and (5) are modeled according to the value of ρ_L with one exception that the anchor and the RIS in the same separate region with the TX are set as LOS channel, e.g., only the direct path and the reflected path in the same region with the TX is set as LOS channel when $\rho_L = 25\%$. The multipath parameter τ_n^l in (2) is modeled as a Poisson point process with arrival rate T_τ, and the parameter α_n^l follows complex Gaussian distribution, i.e., $\mathcal{CN}(0, \Lambda_l)$, with $\{\Lambda_l\}$ being the power delay profile. The power delay profile is modeled according to the simulation process in [19]. For simplicity, we assume that AOA, AOD of NLOS paths in (7) follow uniform distribution, i.e., θ_m^l, $\phi_{m>n}^l \sim U(0, 2\pi), l = 2, \ldots, L$, and path coefficient follows complex Gaussian distribution, i.e., $\delta_m^l \sim \mathcal{CN}(0, \sigma_l^2), l = 2, \ldots, L$.

The localization performance is evaluated using the root mean square error (RMSE), defined by $\mathrm{RMSE} = \sqrt{(1/N_L)\sum_{i=1}^{N_L} \|\tilde{p}_S - p_S\|^2}$, where $\tilde{p}_S$ is the estimated location $\hat{p}_S$ in problem (29) for the coarse localization stage, and p_S^* in problem (40) for the fine localization stage in the ith round experiment, $N_L = 500$ is the total number of experiments.

B. Performance of the Coarse Localization

To demonstrate the performance of the method in the coarse localization stage, we first provide an example of component separation, then show the accuracy of the time-delay-based range estimation, which is accomplished through the peak detection-based TOA estimation method proposed in [21].

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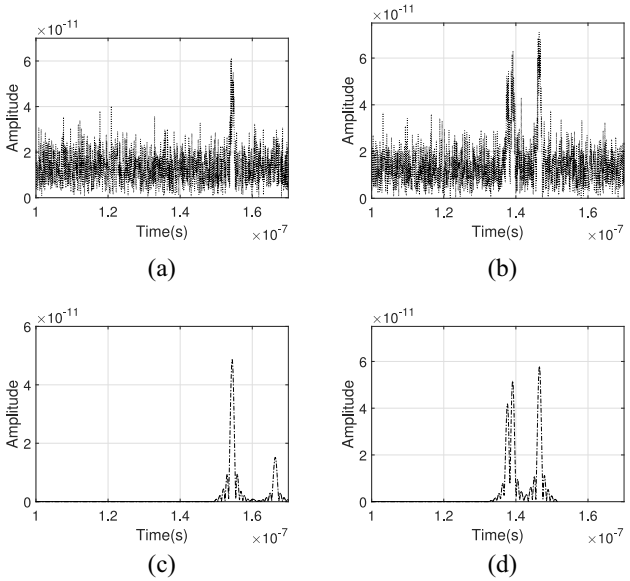


Fig. 2. Example of the extracted and the original component reflected via RIS at noise power $\sigma^2 = -65$ dBm: (a) extracted component in LOS condition; (b) extracted component in NLOS condition; (c) original component in LOS condition; and (d) original component in NLOS condition.

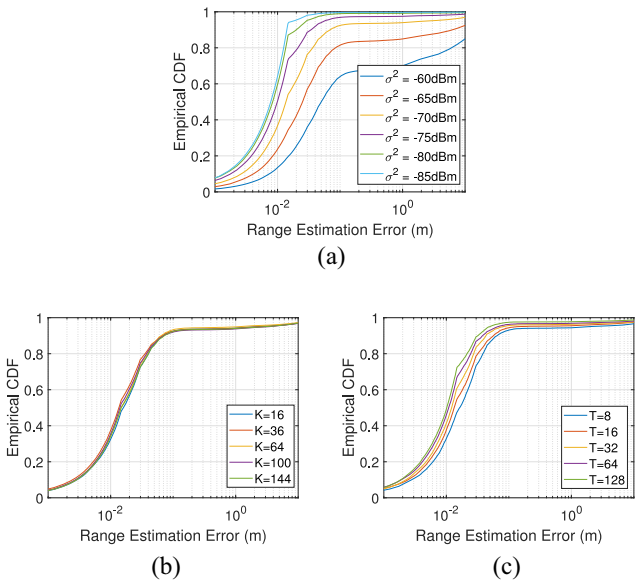


Fig. 3. Empirical CDF of the range estimation error based on the LOS RIS reflected component: (a) different noise power σ^2 with $K = 64$ and T = 8; (b) different number of RIS elements K with $\sigma^2 = -70$ dBm and T = 8; and (c) different length of UWB pulses T with $\sigma^2 = -70$ dBm and $K = 64$.

Finally, we provide the performance of the coarse localization stage.

Fig. 2 shows two examples of the reflected components via RIS. Fig. 2(a) is the reflected component in LOS channel condition extracted from the received signal at noise power $\sigma^2 = -65$ dBm, and Fig. 2(c) is the corresponding ideal signal generated with the simulated LOS channel parameters. Fig. 2(b) is the reflected component in NLOS channel condition extracted from the received signal under the condition of noise power $\sigma^2 = -65$ dBm, and Fig. 2(d) is the corresponding ideal signal generated with the simulated NLOS channel parameters. From Fig. 2, we see that the peaks of the ideal reflected components for both LOS and NLOS conditions are preserved in the separated components with noise. As mentioned in Section III, the peak is crucial for time-delay-based localization.

Fig. 3 illustrates the empirical cumulative distribution function (CDF) of range estimation error, which is based on the LOS reflected component via RIS for various values of noise power σ^2, the length of UWB pulse sequence T, and the number of RIS element K. Fig. 3(a) shows that the CDF exhibits a gradually flattening trend as the noise power σ^2 increases, indicating that the range estimation error progressively increases. Fig. 3(b) indicates that the range estimation error does not vary significantly with the variation in the number of RIS elements K. Fig. 3(c) shows that increasing the length of the UWB pulse sequence can improve range estimation accuracy. From Fig. 3, we see that the noise power σ^2 has a much heavier influence on the range estimation accuracy than the number of RIS elements and the length of UWB pulse sequence.

Fig. 4 shows the localization performance of the coarse localization stage. Fig. 4(a) depicts the RMSE versus the noise power under various conditions of the proportion ρ_L of LOS paths. The figure shows that the RMSE decreases significantly as the noise power decreases. Additionally, the RMSE exhibits a decreasing trend with an increasing ratio of the LOS paths ρ_L. Fig. 4(b) shows the RMSE versus the length of UWB pulse sequence T under various noise conditions. The figure shows that the localization performance is improved by increasing the value of T, which yields a more remarkable improvement on the RMSE when the noise power is larger. When T surpasses a certain threshold, the performance improvement brought by increasing the length of the UWB pulse sequence is relatively weak. From Fig. 4, we observe that a higher proportion of LOS paths, lower noise power levels and larger length of UWB pulse sequence contribute to improved localization performance in the coarse stage.

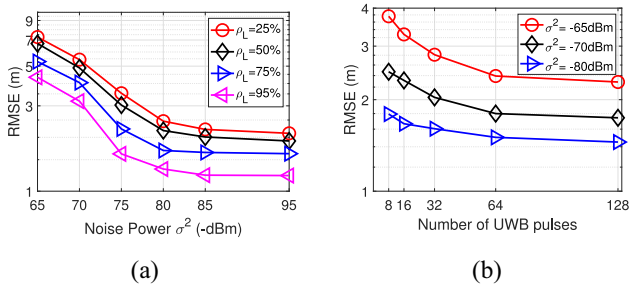
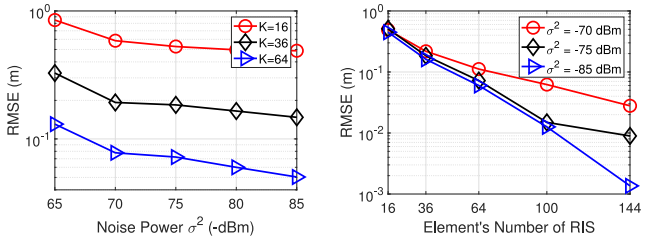


Fig. 4. Localization performance in the coarse localization stage: (a) RMSE versus the noise power σ^2 with different ratio of LOS paths ρ_L and (b) RMSE versus the length of UWB pulses T with $\rho_L = 75\%$.

C. Performance of the Fine Localization

Fig. 5 illustrates the localization performance in the fine localization stage. Fig. 5(a) depicts the RMSE of localization

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(a) (b)

Fig. 5. Localization performance in the fine localization stage: (a) RMSE versus the increasing of noise power σ^2 and (b) RMSE versus the increasing of RIS elements number K.

under various noise levels. The figure shows that the RMSE decreases as the noise power decreases. Fig. 5(b) depicts the RMSE of fine localization by using various RIS with different number of elements. The figure shows that the RMSE decreases as the number of RIS elements increases at various noise power level. As shown in Fig. 5, the noise power has a great influence on the RMSE of the fine localization. We also see that increasing the number of RIS elements can significantly improve the performance of the fine localization.

As shown in Fig. 5, the fine localization stage achieves a significant improvement in accuracy compared to the results in Fig. 4(a) and (b) of the coarse localization stage. Remarkably, a localization accuracy of less than 1 m can be achieved even with a RIS array only containing 16 elements. Furthermore, when the noise level is low, the localization accuracy by using larger RIS arrays can reach as high as 1 mm.

VI. Conclusion

This article addresses the problem of RIS-aided localization in complex NLOS environments. We propose a multistage localization scheme that consists of a coarse and a fine stage for time-delay-based localization. In the coarse localization stage, a T-length UWB pulse sequence transmitted in the mmWave band is used for wireless measurement. A TDOA problem is then formulated based on the extracted components, either reflected via RIS or directly sent to anchors. In the fine localization stage, a reflection response vector estimation process is employed, and the accurate position of the transmitter is obtained by pursuing this reflection vector in a gradient descent search. Simulation results demonstrate that the proposed multistage localization method achieves high performance in complex NLOS environments.

Appendix A
Proof of Real Number $\mathbf{v}_S\hat{\mathbf{\Omega}}_{\hat{n}}^{\star}$

According to (12), $\hat{\mathbf{\Omega}}_{\hat{n}}^{\star}$ has the form

$$\hat{\mathbf{\Omega}}_{\hat{n}}^{\star}=\left(\mathbf{a}^{\mathrm{T}}(\hat{\theta}_{\hat{n}})\mathrm{diag}(\mathbf{a}(\phi_{\hat{n}>\hat{n}}))\right)^{\mathrm{H}} \tag{41}$$

where $\hat{\theta}_{\hat{n}}$ is the estimated AOA based on the coarse estimate $\hat{p}_S$ produced by problem (29) and $\phi_{\hat{n}>\hat{n}}$ is the known AOD from RIS $R_{\hat{n}}$ to anchor $A_{\hat{n}}$.

Substituting $\mathbf{v}_S$ with the definition $\mathbf{v}_S = (\mathbf{a}_r(\kappa_1)\otimes\mathbf{a}_c(\kappa_3))^{\mathrm{T}}\mathrm{diag}(\mathbf{a}(\phi_{\hat{n}>\hat{n}}))$ in Section IV-B, and substituting $\hat{\mathbf{\Omega}}_{\hat{n}}^{\star}$ in (41) into $\mathbf{v}_S\hat{\mathbf{\Omega}}_{\hat{n}}^{\star}$, we have the following equation:

$$\begin{aligned}\mathbf{v}_S\hat{\mathbf{\Omega}}_{\hat{n}}^{\star}&=(\mathbf{a}_r(\kappa_1)\otimes\mathbf{a}_c(\kappa_3))^{\mathrm{T}}\mathbf{\Lambda}\mathbf{\Lambda}^{\mathrm{H}}\left(\bar{\mathbf{a}}_r(\hat{\kappa}_1)\otimes\bar{\mathbf{a}}_c(\hat{\kappa}_3)\right)\\&=(\mathbf{a}_r(\kappa_1)\otimes\mathbf{a}_c(\kappa_3))^{\mathrm{T}}\left(\bar{\mathbf{a}}_r(\hat{\kappa}_1)\otimes\bar{\mathbf{a}}_c(\hat{\kappa}_3)\right)\\&=\mathcal{S}(\mathbf{a}_r(\Delta\kappa_1))\mathcal{S}(\mathbf{a}_c(\Delta\kappa_3))\end{aligned} \tag{42}$$

where $\mathbf{\Lambda}=\mathrm{diag}(\mathbf{a}(\phi_{\hat{n}>\hat{n}}))$ satisfies $\mathbf{\Lambda}\mathbf{\Lambda}^{\mathrm{H}}=\mathbf{I}_K$ because the diagonal elements in $\mathbf{\Lambda}$ are all unit complex, κ_1 and κ_3 are calculated with $\theta_{\hat{n}}$ in (11), and $\hat{\kappa}_1$ and $\hat{\kappa}_3$ are calculated with $\hat{\theta}_{\hat{n}}$ in (11), $\Delta\kappa_1=\kappa_1-\hat{\kappa}_1$, $\Delta\kappa_3=\kappa_3-\hat{\kappa}_3$, $\mathcal{S}(\cdot)$ represents the summation operation to an array.

According to the definition of $\mathbf{a}_r$, $\mathbf{a}_c$ in (9) and (10), we have

$$\begin{aligned}\mathcal{S}(\mathbf{a}_r(\Delta\kappa_1))&=2\sum_{i}^{M_r/2}\cos\left(\frac{\pi d}{\lambda}(M_r+1-2i)\Delta\kappa_1\right)\\\mathcal{S}(\mathbf{a}_c(\Delta\kappa_3))&=2\sum_{i}^{M_c/2}\cos\left(\frac{\pi d}{\lambda}(M_c+1-2i)\Delta\kappa_3\right)\end{aligned} \tag{43}$$

where M_r and M_c are both supposed to be even for simplicity. From (43), we see that $\mathcal{S}(\mathbf{a}_r(\Delta\kappa_1))$ and $\mathcal{S}(\mathbf{a}_c(\Delta\kappa_3))$ are both real numbers, and the inequalities $\mathcal{S}(\mathbf{a}_r(\Delta\kappa_1))\le M_r$ and $\mathcal{S}(\mathbf{a}_c(\Delta\kappa_3))\le M_c$ are satisfied.

Hence, $\mathbf{v}_S\hat{\mathbf{\Omega}}_{\hat{n}}^{\star}$ is a real number and the inequality $\mathbf{v}_S\hat{\mathbf{\Omega}}_{\hat{n}}^{\star}\le K$ is satisfied. That is, the proof.

Appendix B
Partial Derivatives of $L(\kappa_1,\kappa_3)$

The derivative of $L(\kappa_1,\kappa_3)$ w.r.t p_x is

$$\begin{aligned}\frac{\partial L(\kappa_1,\kappa_3)}{\partial p_x}&=\frac{\partial\left(\mathbf{v}(\kappa_1,\kappa_3)-\mathbf{v}_S^*\right)\left(\mathbf{v}(\kappa_1,\kappa_3)-\mathbf{v}_S^*\right)^{\mathrm{H}}}{\partial p_x}\\&=\frac{\partial\left(2K-\mathbf{v}(\kappa_1,\kappa_3)\mathbf{v}_S^{*\mathrm{H}}-\mathbf{v}_S^*\mathbf{v}(\kappa_1,\kappa_3)^{\mathrm{H}}\right)}{\partial p_x}\\&=-\frac{\partial\mathbf{v}(\kappa_1,\kappa_3)}{\partial\kappa_1}\mathbf{v}_S^{*\mathrm{H}}\frac{\partial\kappa_1}{\partial p_x}-\frac{\partial\mathbf{v}(\kappa_1,\kappa_3)}{\partial\kappa_3}\mathbf{v}_S^{*\mathrm{H}}\frac{\partial\kappa_3}{\partial p_x}\\&\quad-\mathbf{v}_S^*\frac{\partial\mathbf{v}(\kappa_1,\kappa_3)^{\mathrm{H}}}{\partial\kappa_1}\frac{\partial\kappa_1}{\partial p_x}-\mathbf{v}_S^*\frac{\partial\mathbf{v}(\kappa_1,\kappa_3)^{\mathrm{H}}}{\partial\kappa_3}\frac{\partial\kappa_3}{\partial p_x}\end{aligned} \tag{44}$$

where

$$\begin{aligned}\frac{\partial\mathbf{v}(\kappa_1,\kappa_3)}{\partial\kappa_1}&=\frac{\partial\mathbf{a}_r^{\mathrm{T}}(\kappa_1)}{\partial\kappa_1}\otimes\mathbf{a}_c^{\mathrm{T}}(\kappa_3)\mathrm{diag}\left(\mathbf{a}(\phi_{\hat{n},\hat{n}})\right)\\\frac{\partial\mathbf{v}(\kappa_1,\kappa_3)}{\partial\kappa_3}&=\mathbf{a}_r^{\mathrm{T}}(\kappa_1)\otimes\frac{\partial\mathbf{a}_c^{\mathrm{T}}(\kappa_3)}{\partial\kappa_3}\mathrm{diag}\left(\mathbf{a}(\phi_{\hat{n},\hat{n}})\right)\\\frac{\partial\mathbf{v}(\kappa_1,\kappa_3)^{\mathrm{H}}}{\partial\kappa_1}&=\left\{\mathrm{diag}\left(\mathbf{a}(\phi_{\hat{n},\hat{n}})\right)\right\}^{\mathrm{H}}\frac{\partial\bar{\mathbf{a}}_r(\kappa_1)}{\partial\kappa_1}\otimes\bar{\mathbf{a}}_c(\kappa_3)\\\frac{\partial\mathbf{v}(\kappa_1,\kappa_3)^{\mathrm{H}}}{\partial\kappa_3}&=\left\{\mathrm{diag}\left(\mathbf{a}(\phi_{\hat{n},\hat{n}})\right)\right\}^{\mathrm{H}}\bar{\mathbf{a}}_r(\kappa_1)\otimes\frac{\partial\bar{\mathbf{a}}_c(\kappa_3)}{\partial\kappa_3}.\end{aligned}$$

The partial $(\partial\kappa_1/\partial p_x)$ and $(\partial\kappa_3/\partial p_x)$ in (44) are expressed based on the RIS $R_{\hat{n}}$'s coordination $\mathcal{R}_{\hat{n}}$, as shown in Fig. 1(b).

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APPENDIX C
EFFECTIVENESS ANALYSIS OF THE GRADIENT DESCENT

Suppose $p_S^\star$ is the optimal solution of problem (40), then the objective vector $\mathbf{v}_S^*$ can be expressed in

$$\mathbf{v}_S^\star = \left(\mathbf{a}_r(\kappa_1^\star) \otimes \mathbf{a}_c(\kappa_3^\star)\right)^{\mathrm{T}} \mathrm{diag}\left(\mathbf{a}(\phi_{\hat{n},\hat{n}})\right) \tag{45}$$

where $\kappa_1^\star$ and $\kappa_3^\star$ is the variable constructed using $p_S^\star$ in (40).

First, we demonstrate that the uniqueness of the optimal solution $p_S^\star$ is satisfied.

If another optimal solution $\tilde{p}_S$ exists, then $\tilde{\kappa}_1$ and $\tilde{\kappa}_3$ constructed by $\tilde{p}_S$ also satisfies (45). Defining the bias as $\Delta p = \tilde{p}_S - p_S^\star$, we have $\Delta\kappa_1 = \tilde{\kappa}_1 - \kappa_1^\star$, and $\Delta\kappa_3 = \tilde{\kappa}_3 - \kappa_3^\star$. We derive $L(\tilde{\kappa}_1, \tilde{\kappa}_3)$ by substituting $\mathbf{v}_S^\star$ in (45) as follows:

$$\begin{aligned} L(\tilde{\kappa}_1, \tilde{\kappa}_3) &= \left(\mathbf{v}(\tilde{\kappa}_1, \tilde{\kappa}_3) - \mathbf{v}_S^\star\right)\left(\mathbf{v}(\tilde{\kappa}_1, \tilde{\kappa}_3) - \mathbf{v}_S^\star\right)^{\mathrm{H}} \\ &= 2K - \mathbf{v}(\tilde{\kappa}_1, \tilde{\kappa}_3)\mathbf{v}_S^{\star\mathrm{H}} - \mathbf{v}_S^\star \mathbf{v}(\tilde{\kappa}_1, \tilde{\kappa}_3)^{\mathrm{H}} \\ &= 2K - \left(\mathbf{a}_r(\tilde{\kappa}_1) \otimes \mathbf{a}_c(\tilde{\kappa}_3)\right)^{\mathrm{T}} \Lambda\Lambda^{\mathrm{H}} \left(\bar{\mathbf{a}}_r(\kappa_1^\star) \otimes \bar{\mathbf{a}}_c(\kappa_3^\star)\right) \\ &\quad - \left(\mathbf{a}_r(\kappa_1^\star) \otimes \mathbf{a}_c(\kappa_3^\star)\right)^{\mathrm{T}} \Lambda\Lambda^{\mathrm{H}} \left(\bar{\mathbf{a}}_r(\tilde{\kappa}_1) \otimes \bar{\mathbf{a}}_c(\tilde{\kappa}_3)\right) \\ &= 2K - 2\mathcal{S}(\mathbf{a}_r(\Delta\kappa_1))\mathcal{S}(\mathbf{a}_c(\Delta\kappa_3)). \end{aligned} \tag{46}$$

Based on the expression of $\mathcal{S}(\cdot)$ in (43), it is observed that in order for $L(\tilde{\kappa}_1, \tilde{\kappa}_3) = 0$, the conditions $\Delta\kappa_1 = 0$ and $\Delta\kappa_3 = 0$ must be satisfied. As a result, $\tilde{p}_S = p_S^\star$ is also satisfied for the trigonometric expression of azimuth and elevation shown in Fig. 1(b).

Second, we demonstrate that problem (40) is convex in the near domain of its optimal solution $p_S^\star$.

We approximate the geometric relationship between κ_1, κ_3 and p_S as linear functions in the feasible solution domain near $p_S^\star$. In this case, the complexity of convexity proof is reduced to prove the following Hessian matrix semi-definite positive:

$$\begin{bmatrix} \frac{\partial^2 L(\kappa_1,\kappa_3)}{\partial \kappa_1^2} & \frac{\partial^2 L(\kappa_1,\kappa_3)}{\partial \kappa_1 \partial \kappa_3} \\ \frac{\partial^2 L(\kappa_1,\kappa_3)}{\partial \kappa_1 \partial \kappa_3} & \frac{\partial^2 L(\kappa_1,\kappa_3)}{\partial \kappa_3^2} \end{bmatrix} \succeq 0. \tag{47}$$

Supposing M_r and M_c are even, and substituting $\mathbf{v}_S^\star$ in (45) into $L(\kappa_1, \kappa_3)$, we have the expression of these second-order differential items with $\Delta\kappa_1$ and $\Delta\kappa_3$ as follows:

$$\begin{aligned} \frac{\partial^2 L(\kappa_1,\kappa_3)}{\partial \kappa_1^2} &= 8 \sum_{i}^{M_c/2} \cos\left(\frac{\pi d}{\lambda}(M_c + 1 - 2i)\Delta\kappa_3\right) \\ &\quad \times \sum_{i=1}^{M_r/2} \left(\frac{\pi^2 d^2}{\lambda^2}(M_r + 1 - 2i)^2 \cos\left(\frac{\pi d}{\lambda}(M_r + 1 - 2i)\Delta\kappa_1\right)\right) \\ \frac{\partial^2 L(\kappa_1,\kappa_3)}{\partial \kappa_3^2} &= 8 \sum_{i}^{M_r/2} \cos\left(\frac{\pi d}{\lambda}(M_r + 1 - 2i)\Delta\kappa_1\right) \\ &\quad \times \sum_{i=1}^{M_c/2} \left(\frac{\pi^2 d^2}{\lambda^2}(M_c + 1 - 2i)^2 \cos\left(\frac{\pi d}{\lambda}(M_c + 1 - 2i)\Delta\kappa_3\right)\right) \\ \frac{\partial^2 L(\kappa_1,\kappa_3)}{\partial \kappa_1 \partial \kappa_3} &= 4 \sum_{i}^{M_c/2} \cos\left(\frac{\pi d}{\lambda}(M_c + 1 - 2i)\Delta\kappa_3\right) \\ &\quad \times \sum_{i=1}^{M_r/2} \left(\frac{\pi d}{\lambda}(M_r + 1 - 2i) \sin\left(\frac{\pi d}{\lambda}(M_r + 1 - 2i)\Delta\kappa_1\right)\right) \\ &\quad + 4 \sum_{i}^{M_r/2} \cos\left(\frac{\pi d}{\lambda}(M_r + 1 - 2i)\Delta\kappa_1\right) \\ &\quad \times \sum_{i=1}^{M_c/2} \left(\frac{\pi d}{\lambda}(M_c + 1 - 2i) \sin\left(\frac{\pi d}{\lambda}(M_c + 1 - 2i)\Delta\kappa_3\right)\right) \end{aligned}$$

where $\Delta\kappa_1 = \kappa_1 - \kappa_1^\star$ and $\Delta\kappa_3 = \kappa_3 - \kappa_3^\star$. We see that $[\partial^2 L(\kappa_1, \kappa_3)/\partial\kappa_1^2][\partial^2 L(\kappa_1, \kappa_3)/\partial\kappa_3^2] - [\partial^2 L(\kappa_1, \kappa_3)/\partial\kappa_1\partial\kappa_3][\partial^2 L(\kappa_1, \kappa_3)/\partial\kappa_1\partial\kappa_3] \geq 0$ can be satisfied when $\Delta\kappa_1$ and $\Delta\kappa_3$ are sufficiently small.

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